25. Low Power Asynchronous General Purpose Timer (AGT)

25.1 Overview

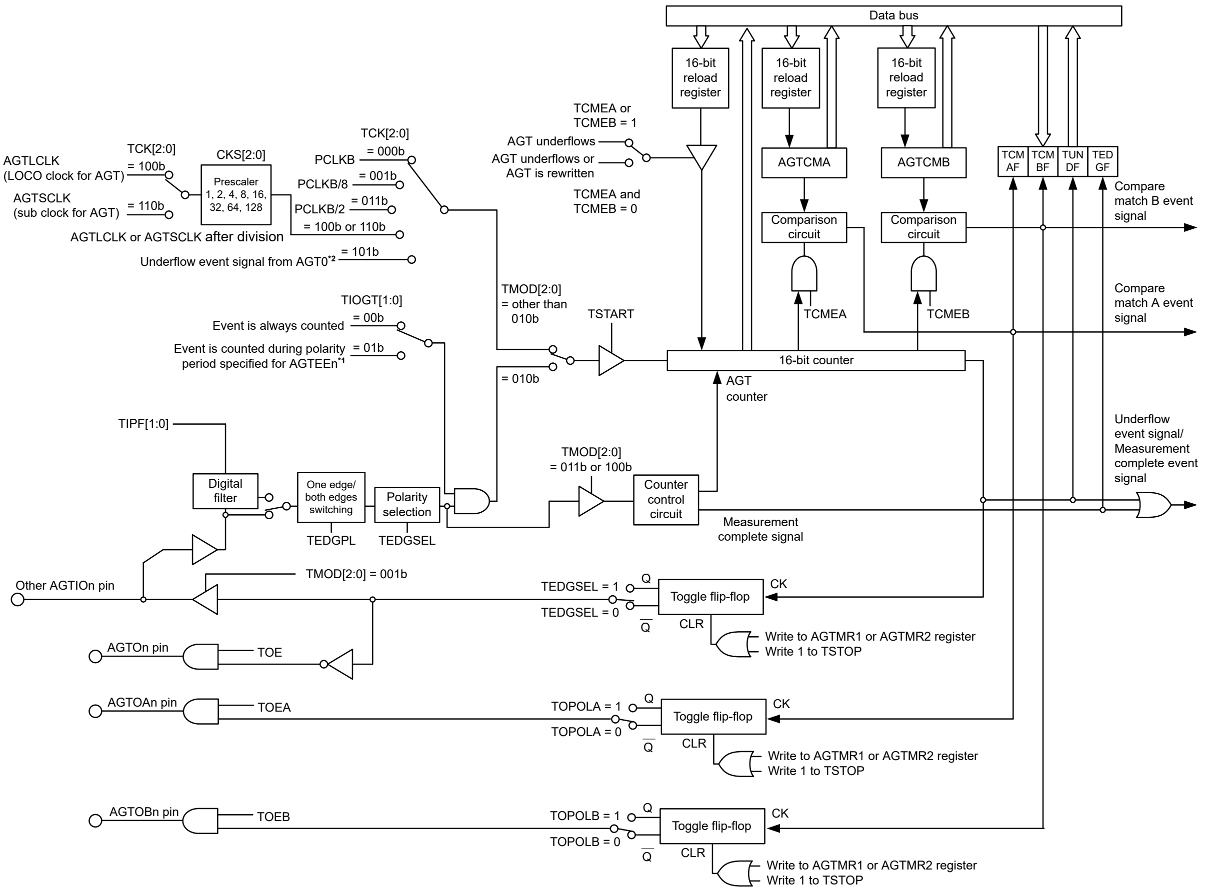
The Low Power Asynchronous General Purpose Timer (AGT) is a 16-bit timer that can be used for pulse output, external pulse width or period measurement, and counting external events. This timer consists of a reload register and a down counter. The reload register and the down counter are allocated to the same address, and can be accessed with the AGT register.

Table 25.1 lists the AGT specifications, Figure 25.1 shows a block diagram, and Table 25.2 lists the I/O pins.

Table 25.1 AGT specifications

Parameter		Description
Operating modes	Timer mode	The count source is counted
	Pulse output mode	The count source is counted and the output is inverted at each timer underflow
	Event counter mode	An external event is counted
	Pulse width measurement mode	An external pulse width is measured
	Pulse period measurement mode	An external pulse period is measured
Number of Channels		16 bits × 2 channels (AGTn (n = 0, 1))
Count source (operating clock)*2	Timer mode	PCLKB, PCLKB/2, PCLKB/8, AGTLCLK/d, AGTSCLK/d (d = 1, 2, 4, 8, 16, 32, 64, or 128), or underflow signal of AGT0 selectable.*1
	Pulse output mode	
	Pulse width measurement mode	
	Pulse period measurement mode	
	Event counting mode	External event input
Interrupt and Event Link function		• Underflow event signal or measurement complete event signal – When the counter underflows – When the measurement of the active width of the external input pin (AGTIOn) completes in pulse width measurement mode – When the set edge of the external input pin (AGTIOn) is input in pulse period measurement mode. • Compare match A event signal – When the values of AGT register and AGTCMA register matched (compare match A function enabled). • Compare match B event signal – When the values of AGT and AGTCMB registers matched (compare match B function enabled). • Return from Software Standby mode can be performed with AGT1_AGTI, AGT1_AGTCMAI, or AGT1_AGTCMBI*3
Selectable functions		• Compare match function One or two of the AGT Compare Match A register and AGT Compare Match B register is selectable.
Module-stop function		Module-stop state can be set for each channel to reduce power consumption.
TrustZone Filter		Security and Privilege attribution can be set for each channel.

Note 1. AGT0 cannot use underflow signal. AGT1 connects directly with the underflow event signal from the AGT0 timer.
Note 2. Satisfy the frequency of the peripheral module clock (PCLKB) ≥ the frequency of the count source clock. When selecting AGTLCLK/d or AGTSCLK/d as the count source clock, the frequency of PCLKB should be at least twice that of the count source clock.
Note 3. For details, see section 11, Low Power Mode.



Note 1. The polarity can be selected by the EEPS bit in the AGTISR register.
Note 2. AGT0 cannot use AGT0 underflow event. AGT1 uses the underflow of AGT0.

Figure 25.1 AGT block diagram

Table 25.2 AGT I/O pins

Pin name	I/O	Function
AGTEEn	Input	External event input enable for AGT
AGTIOn	Input/output	External event input and pulse output for AGT
AGTOn	Output	Pulse output for AGT
AGTOAn	Output	Compare match A output for AGT
AGTOBn	Output	Compare match B output for AGT

Note: Channel number: n = 0, 1

25.2 Register Descriptions

25.2.1 AGT : AGT Counter Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x00

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
15:0	n/a	16-bit counter and reload register Setting range : 0x0000 to 0xFFFF	R/W

Note: S-TYPE-3, P-TYPE-3

AGTn.AGT is a 16-bit register. The write value is written to the reload register and the read value is read from the counter.

The states of the reload register and the counter change according to the TSTART bit in the AGTCR register and TCMEA/TCMEB bit in the AGTCMSR register. For details, see section 25.3.1. Reload Register and Counter Rewrite Operation.

When 1 is written to the TSTOP bit in the AGTCR register, AGT counter is forcibly stopped and set to 0xFFFF.

When the TCK[2:0] bits setting in the AGTMR1 register are a value other than 001b (PCLKB/8) or 011b (PCLKB/2), if the AGT register is set to 0x0000, a request signal to the ICU, the DTC, the DMAC, and the ELC is generated once immediately after the count starts. The AGTOn, AGTIOn pin output are toggled.

When the AGT register is set to 0x0000 in event counter mode, regardless of the value of TCK[2:0] bits, a request signal to the ICU, the DTC, the DMAC, and the ELC is generated once immediately after the count starts.

In addition, the AGTOn pin output is toggled even during a period other than the specified count period. When the AGT register is set to 0x0001 or more, a request signal is generated each time AGT underflows.

25.2.2 AGTCMA : AGT Compare Match A Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x02

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
15:0	n/a	16-bit compare match A data is stored.[*1] Setting range : 0x0000 to 0xFFFF	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Set the AGTCMA register to 0xFFFF when compare match A is not used.

The AGTCMA register is a read/write register to set a value for compare match with the AGT counter. The states of the reload register and compare register A change according to the TSTART bit in the AGTCR register. For details, see section 25.3.2. Reload Register and AGT Compare Match A/B Register Rewrite Operation.

25.2.3 AGTCMB : AGT Compare Match B Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
15:0	n/a	16-bit compare match B data is stored.[*1] Setting range : 0x0000 to 0xFFFF	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Set the AGTCMB register to 0xFFFF when compare match B is not used.

The AGTCMB register is a read/write register to set a value for compare match with the AGT counter. The states of the reload register and compare register B change according to the TSTART bit in the AGTCR register. For details, see section 25.3.2. Reload Register and AGT Compare Match A/B Register Rewrite Operation.

25.2.4 AGTCR : AGT Control Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TCMBF	TCMAF	TUNDF	TEDGF	—	TSTOP	TCSTF	TSTART
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSTART	AGT Count Start*2 0: Count stops 1: Count starts	R/W
1	TCSTF	AGT Count Status Flag*2 0: Count stopped 1: Count in progress	R
2	TSTOP	AGT Count Forced Stop*1 0: Writing is invalid 1: The count is forcibly stopped	W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TEDGF	Active Edge Judgment Flag 0: No active edge received 1: Active edge received	R/(W)*3
5	TUNDF	Underflow Flag 0: No underflow 1: Underflow	R/(W)*3
6	TCMAF	Compare Match A Flag 0: No match 1: Match	R/(W)*3
7	TCMBF	Compare Match B Flag 0: No match 1: Match	R/(W)*3

Note: S-TYPE-3, P-TYPE-3

Note 1. When 1 (count is forcibly stopped) is written to the TSTOP bit, the TSTART bit and TCSTF flag are initialized at the same time. The pulse output level is also initialized. The read value is 0.

Note 2. For information on using the TSTART bit and TCSTF flag, see section 25.4.1. Count Operation Start and Stop Control.

Note 3. Only 0 can be written to clear the flag.

TSTART bit (AGT Count Start)

The count operation is started by writing 1 to the TSTART bit and stopped by writing 0. When the TSTART bit is set to 1 (count starts), the TCSTF flag is set to 1 (count in progress) in synchronization with the count source. Also, after 0 is written to the TSTART bit, the TCSTF flag is set to 0 (count stops) in synchronization with the count source. For details, see section 25.4.1. Count Operation Start and Stop Control.

TCSTF flag (AGT Count Status Flag)

The TCSTF flag indicates the AGT count status.

[Setting condition]

- When 1 is written to the TSTART bit (the TCSTF flag is set to 1 in synchronization with the count source).

[Clearing conditions]

- When 0 is written to the TSTART bit (the TCSTF flag is set to 0 in synchronization with the count source)
- When 1 is written to the TSTOP bit.

TSTOP bit (AGT Count Forced Stop)

When 1 is written to the TSTOP bit, the count is forcibly stopped. The read value is 0.

TEDGF flag (Active Edge Judgment Flag)

The TEDGF flag indicates that an active edge was detected.

[Setting condition]

- When the measurement of the active width of the external input pin (AGTIOn) is complete in pulse width measurement mode
- When the set edge of the external input pin (AGTIOn) is input in pulse period measurement mode.

[Clearing condition]

- When 0 is written to this flag by software.

TUNDF flag (Underflow Flag)

The TUNDF flag indicates that the counter underflowed.

[Setting condition]

- When the counter underflows.

[Clearing condition]

- When 0 is written to this flag by software.

TCMAF flag (Compare Match A Flag)

The TCMAF flag indicates that compare match A was detected.

[Setting condition]

- When the value in the AGT register matches the value in the AGTCMA register.

[Clearing condition]

- When 0 is written to this flag by software.

TCMBF flag (Compare Match B Flag)

The TCMBF flag indicates that compare match B was detected.

[Setting condition]

- When the value in the AGT register matches the value in the AGTCMB register.

[Clearing condition]

- When 0 is written to this flag by software.

25.2.5 AGTMR1 : AGT Mode Register 1

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x09

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TCK[2:0]			TEDGPL	TMOD[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	TMOD[2:0]	Operating Mode*3 0 0 0: Timer mode 0 0 1: Pulse output mode 0 1 0: Event counter mode 0 1 1: Pulse width measurement mode 1 0 0: Pulse period measurement mode Others: Setting prohibited	R/W
3	TEDGPL	Edge Polarity*4 0: Single-edge 1: Both-edge	R/W
6:4	TCK[2:0]	Count Source*1 *2 *5 *7 0 0 0: PCLKB 0 0 1: PCLKB/8 0 1 1: PCLKB/2 1 0 0: Divided clock AGTLCLK specified by CKS[2:0] bits in the AGTMR2 register 1 0 1: Underflow event signal from AGT0*6 1 1 0: Divided clock AGTSCLK specified by CKS[2:0] bits in the AGTMR2 register Others: Setting prohibited	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Write access to the AGTMR1 register initializes the output from the AGTOn, AGTIOn, AGTOAn, and AGTOBn pins. For details on the output level at initialization, see section 25.2.7. AGTIOC : AGT I/O Control Register.

Note 1. When event counter mode is selected, the external input pin (AGTIOn) is selected as the count source regardless of the setting of TCK[2:0] bits.

Note 2. Do not switch count sources during count operation. Only switch count sources when both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count stops).

Note 3. The operating mode can only be changed when the count is stopped while both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count is stopped). Do not change the operating mode during count operation.

Note 4. The TEDGPL bit is enabled only in event counter mode.

Note 5. To run AGT in Software Standby mode, select AGTLCLK or AGTSCLK (TCK[2:0] = 100b, 110b).

Note 6. AGT0 cannot use AGT0 underflow (setting prohibited). AGT1 uses the AGT0 underflow.

Note 7. Do not change the TCK[2:0] bits when the CKS[2:0] bits in the AGTMR2 register is not 000b. First, change the CKS[2:0] bits in the AGTMR2 register to 000b. Then change the TCK[2:0] bits and wait for one cycle of the count source.

25.2.6 AGTMR2 : AGT Mode Register 2

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LPM	—	—	—	—	CKS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CKS[2:0]	AGTLCLK or AGTSCLK Count Source Clock Frequency Division Ratio*1 *2 *3 0 0 0: 1/1 0 0 1: 1/2 0 1 0: 1/4 0 1 1: 1/8 1 0 0: 1/16 1 0 1: 1/32 1 1 0: 1/64 1 1 1: 1/128	R/W
6:3	—	These bits are read as 0. The write value should be 0.	R/W
7	LPM	Low Power Mode 0: Normal mode 1: Low power mode	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite the CKS[2:0] bits during count operation. Only rewrite the CKS[2:0] bits when both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count stops).
Note 2. When count source is AGTLCLK or AGTSCLK, the switch of CKS[2:0] bits is valid.
Note 3. Do not switch the TCK[2:0] bits in the AGTMR1 register when CKS[2:0] bits are not 000b. Switch the TCK[2:0] bits in the AGTMR1 register after CKS[2:0] bits are set to 000b, and wait for 1 cycle of the count source.

CKS[2:0] bit (AGTLCLK or AGTSCLK Count Source Clock Frequency Division Ratio)

CKS[2:0] bits select the Count Source Clock Frequency Division Ratio for AGTLCLK or AGTSCLK.

LPM bit (Low Power Mode)

The LPM bit sets the low power operation, which impacts access to certain AGT registers. Set this bit to 1 to operate in low power.

When this bit is 1, access to the following registers is prohibited:

- AGT/AGTCMA/AGTCMB/AGTCR.

After this bit is switched from 1 to 0, the first access to the register is constrained as follows:

- When reading from the AGT register, read AGT register twice. Only the second reading of data is valid.
- When writing to the AGT, AGTCMA, AGTCMB, and AGTCR register, allow at least 2 cycles of the count source clock when writing to the register.
- When confirm the value written to the AGT, AGTCMA, AGTCMB, and AGTCR registers.
  - When the count operation is stopped; after writing data, it can be read in the next cycle.
  - When the count operation is operating; after writing data, it can be read 4 cycles after the count source clock.

Figure 25.2 shows the flow of how to write LPM bit

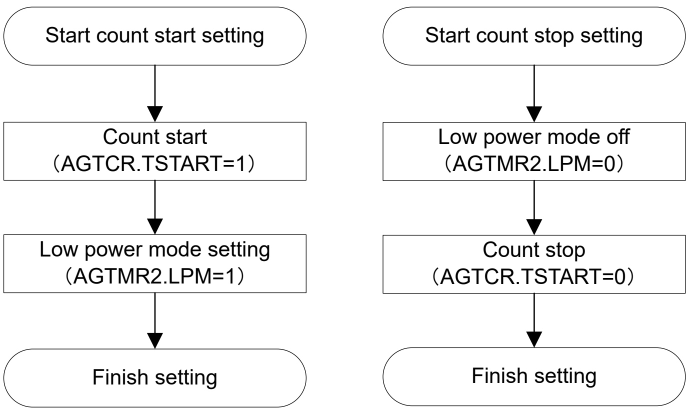


Figure 25.2 LPM how to write flow chart

25.2.7 AGTIOC : AGT I/O Control Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TIOGT[1:0]		TIPF[1:0]		—	TOE	—	TEDGSEL
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TEDGSEL	I/O Polarity Switch Function varies depending on the operating mode (see Table 25.3 and Table 25.4).	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	TOE	AGTOn pin Output Enable 0: AGTOn pin output disabled 1: AGTOn pin output enabled	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TIPF[1:0]	Input Filter*3 These bits specify the sampling frequency of the filter for the AGTIOn input. If the input to the AGTIOn pin is sampled and the value matches three successive times, that value is taken as the input value. 0 0: No filter 0 1: Filter sampled at PCLKB 1 0: Filter sampled at PCLKB/8 1 1: Filter sampled at PCLKB/32	R/W
7:6	TIOGT[1:0]	Count Control*1 *2 0 0: Event is always counted 0 1: Event is counted during polarity period specified for AGTEEn pin Others: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. When AGTEEn pin is used, the polarity to count an event can be selected with the EEPS bit in the AGTISR register.
Note 2. TIOGT[1:0] bits are enabled only in event counter mode.
Note 3. When event counter mode operation is performed during Software Standby mode, the digital filter function cannot be used.

TEDGSEL bit (I/O Polarity Switch)

The TEDGSEL bit switches the AGTOn pin output polarity and the AGTIOn pin input/output edge and polarity.

In pulse output mode, it only controls polarity of the AGTOn pin output and AGTIOn pin output. AGTOn pin output and AGTIOn pin output are initialized when the AGTMR1 register is written or the TSTOP bit in the AGTCR register is written with 1.

TOE bit (AGTOn pin Output Enable)

The TOE bit selects whether the AGTOn pin output is disabled or enabled.

TIPF[1:0] bits (Input Filter)

The TIPF[1:0] bits specify the sampling frequency of the AGTIOn pin input filter. When the input to the AGTIOn pin is sampled and the values match three times in succession, the value is regarded as the input value.

TIOGT[1:0] bits (Count Control)

The TIOGT[1:0] bits control the event count.

Table 25.3 AGTIOn pin I/O edge and polarity switching

Operating mode	Function
Timer mode	Not used
Pulse output mode	0: Output is started at high (initialization level: high) i.e. inverted output 1: Output is started at low (initialization level: low). i.e. normal output
Event counter mode	0: Count on rising edge 1: Count on falling edge.
Pulse width measurement mode	0: Low-level width is measured 1: High-level width is measured.
Pulse period measurement mode	0: Measure from one rising edge to the next rising edge 1: Measure from one falling edge to the next falling edge.

Table 25.4 AGTOn pin output polarity switching

Operating mode	Function
All modes	0: Output is started at low (initial level: low): Normal output 1: Output is started at high (initial level: high): Inverted output

25.2.8 AGTISR : AGT Event Pin Select Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	EEPS	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
2	EEPS	AGTEEn Polarity Selection 0: An event is counted during the low-level period 1: An event is counted during the high-level period	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

EEPS bit (AGTEEn Polarity Selection)

The EEPS bit selects the polarity of events to be counted.

25.2.9 AGTCMSR : AGT Compare Match Function Select Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TOPOLB	TOEB	TCMEB	—	TOPOLA	TOEA	TCMEA
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCMEA	AGT Compare Match A Register Enable*1 *2 0: AGT Compare match A register disabled 1: AGT Compare match A register enabled	R/W
1	TOEA	AGTOAn Pin Output Enable*1 *2 0: AGTOAn pin output disabled 1: AGTOAn pin output enabled	R/W
2	TOPOLA	AGTOAn Pin Polarity Select*1 *2 0: AGTOAn pin output is started on low. i.e. normal output 1: AGTOAn pin output is started on high. i.e. inverted output	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TCMEB	AGT Compare Match B Register Enable*1 *2 0: Compare match B register disabled 1: Compare match B register enabled	R/W
5	TOEB	AGTOBn Pin Output Enable*1 *2 0: AGTOBn pin output disabled 1: AGTOBn pin output enabled	R/W

Bit	Symbol	Function	R/W
6	TOPOLB	AGTOBn Pin Polarity Select*1 *2 0: AGTOBn pin output is started on low. i.e. normal output 1: AGTOBn pin output is started on high. i.e. inverted output	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Do not rewrite the AGTCMSR register during a count operation. Only rewrite the AGTCMSR register when both the TSTART bit and TCSTF flag in the AGTCR register are set to 0 (count stops).

Note 2. Do not set 1 when in pulse width measurement mode or pulse period measurement mode.

25.2.10 AGTIOSEL : AGT Pin Select Register

Base address: AGTn = 0x4022_1000 + 0x0100 × n (n = 0, 1)
AGTn_NS = 0x5022_1000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TIES	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	TIES	AGTIOn Pin Input Enable 0: External event input is disabled during Software Standby mode 1: External event input is enabled during Software Standby mode	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The AGTIOSEL register sets the AGTIOn pin when using the AGTIOn pin in Software Standby mode.

TIES bit (AGTIOn Pin Input Enable)

The TIES bit enables or disables an external event input.

25.3 Operation

25.3.1 Reload Register and Counter Rewrite Operation

Regardless of the operating mode, the timing of the rewrite operation to the reload register and the counter differs depending on the value of the TSTART bit in the AGTCR register and of the TCMEA or TCMEB bit in the AGTCMSR register.
When the TSTART bit is 0 (count stops), the count value is directly written to the reload register and the counter. When the TSTART bit is 1 (count starts) and the TCMEA bit and TCMEB bit are 0 (AGT compare match A/B register are invalid), the value is written to the reload register in synchronization with the count source, and then to the counter in synchronization with the next count source. When the TSTART bit is 1 (count starts) and the TCMEA bit or the TCMEB bit is 1 (AGT compare match A register or compare match B register is valid), the value is written to the reload register in synchronization with the count source, and then to the counter in synchronization with the underflow of the counter.

Figure 25.3 and Figure 25.4 show the timing of rewrite operation with TSTART bit value and TCMEA/TCMEB bit value.

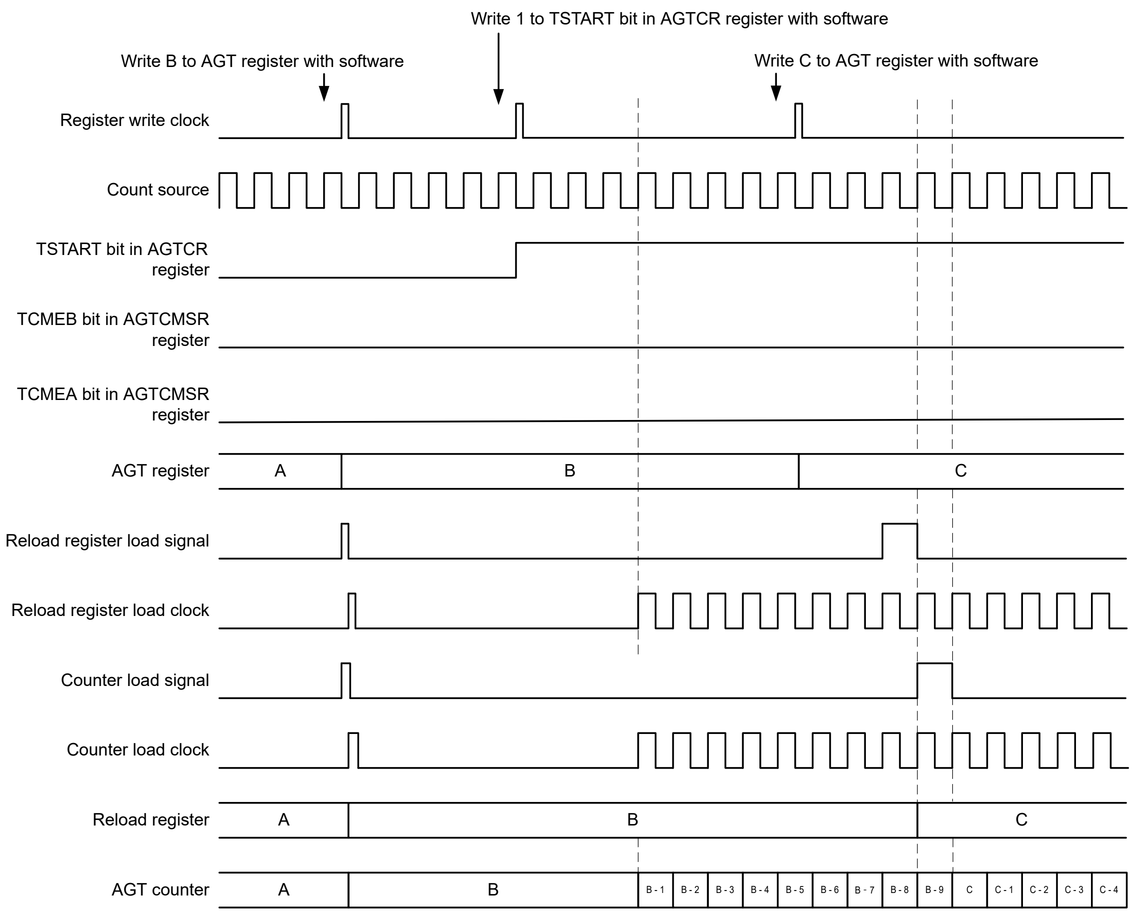


Figure 25.3 Timing of rewrite operation with TSTART, TCMEA, and TCMEB bit value when AGT compare match A register and AGT compare match B register is invalid

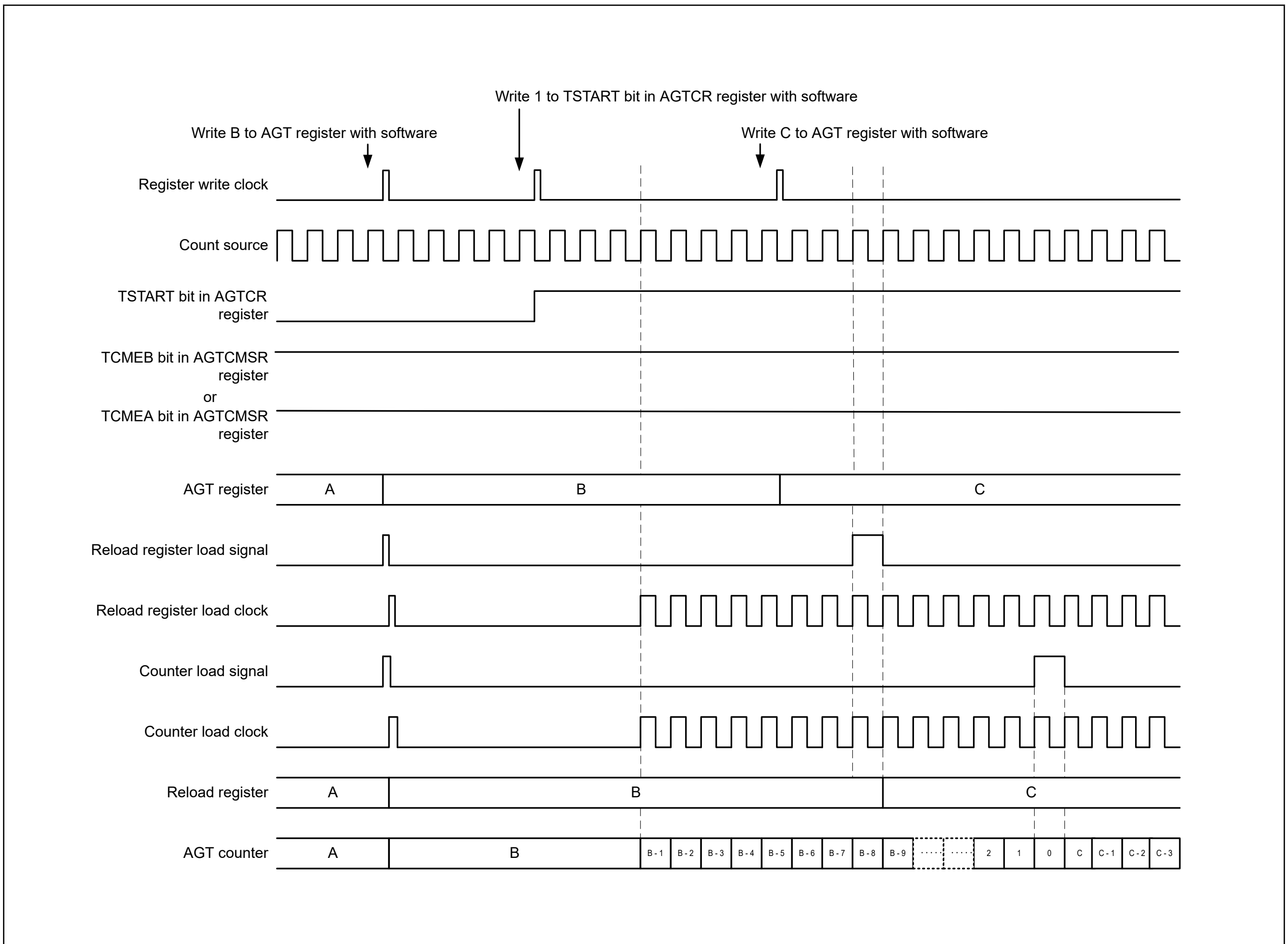


Figure 25.4 Timing of rewrite operation with TSTART bit value and TCMEA or TCMEB bit value when AGT compare match A register or AGT compare match B register is valid

25.3.2 Reload Register and AGT Compare Match A/B Register Rewrite Operation

Regardless of the operating mode, the timing of the rewrite operation to the reload register and AGT compare register A/B depends on the value of the TSTART bit in the AGTCR register. When the TSTART bit is 0 (count stops), the count value is directly written to the reload register and AGT compare register A/B. When the TSTART bit is 1 (count starts), the value is written to the reload register in synchronization with the count source, and then to the compare register in synchronization with the underflow of the counter.

Figure 25.5 shows the timing of rewrite operation with TSTART bit value for compare register A. AGT Compare register B is of the same timing as AGT compare register A.

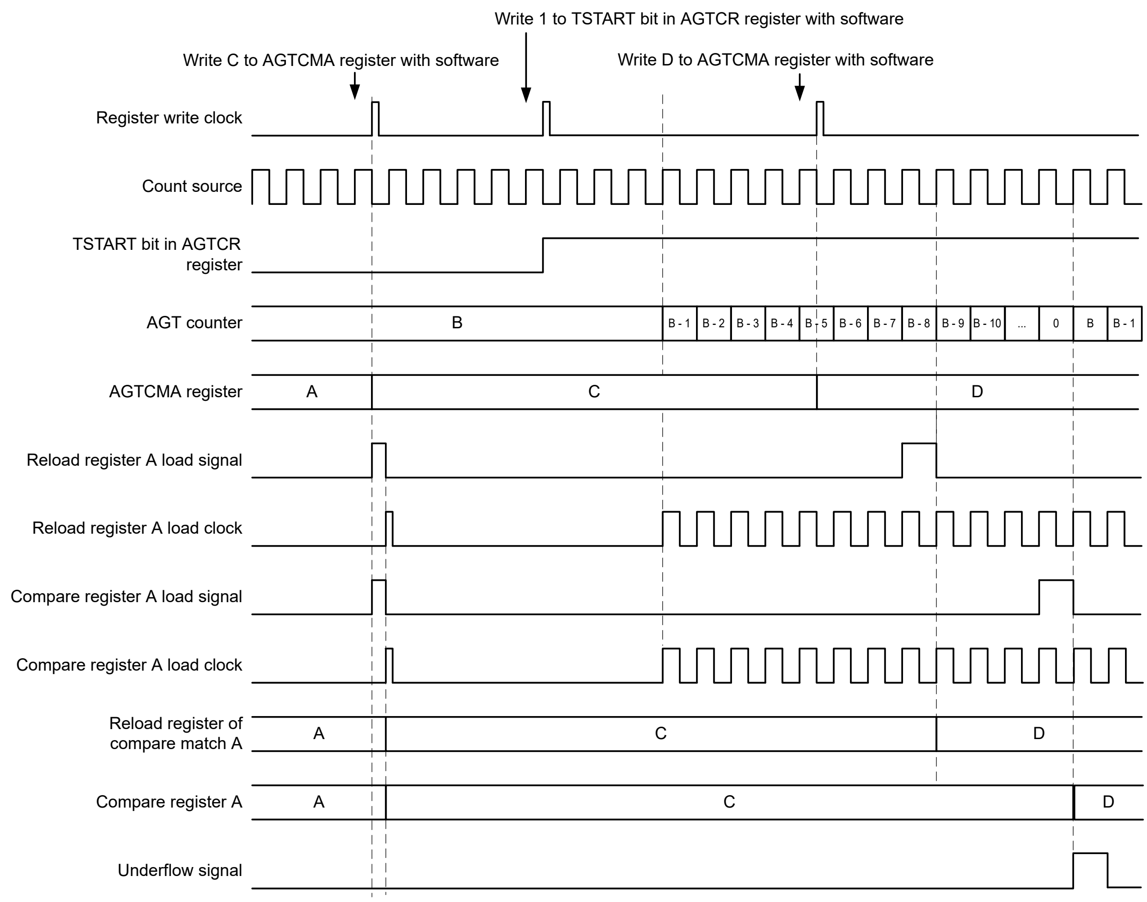


Figure 25.5 Timing of rewrite operation with the TSTART bit value for AGT compare register A

25.3.3 Timer Mode

In this mode, the AGT counter is decremented by the count source selected with the TCK[2:0] bits in the AGTMR1 register. In timer mode, the count value is decremented by 1 on each rising edge of the count source. When the count value reaches 0x0000 and the next count source is input, an underflow occurs, and an interrupt request is generated.

Figure 25.6 shows the operation example in timer mode.

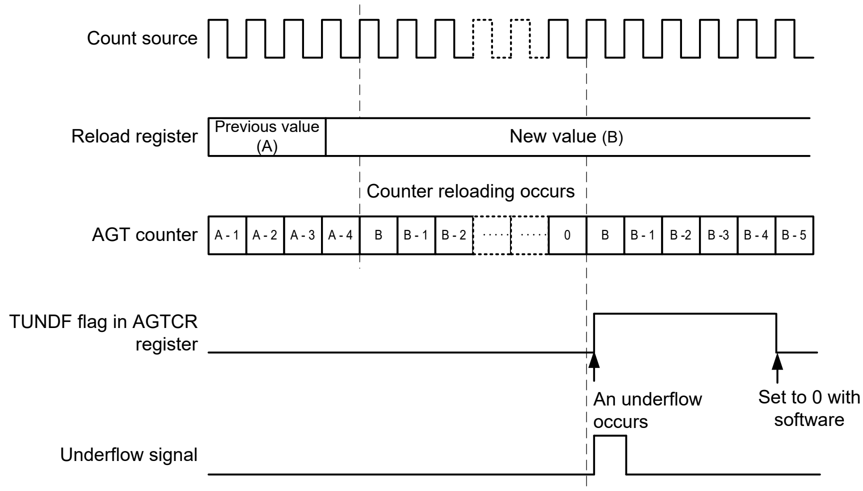


Figure 25.6 Operation example in timer mode

25.3.4 Pulse Output Mode

In pulse output mode, the counter is decremented by the count source selected with the TCK[2:0] bits in the AGTMR1 register, and the output level of the AGTIOn and AGTOn pins inverted each time an underflow occurs.

In pulse output mode, the count value is decremented by 1 on each rising edge of the count source. When the count value reaches 0x0000 and the next count source is input, an underflow occurs, and an interrupt request is generated. In addition, a pulse can be output from the AGTIOn and AGTOn pins. The output level is inverted each time an underflow occurs. The pulse output from the AGTOn pin can be stopped with the TOE bit in the AGTIOC register. The output level can be selected with the TEDGSEL bit in the AGTIOC register.

Figure 25.7 shows the operation example in pulse output mode.

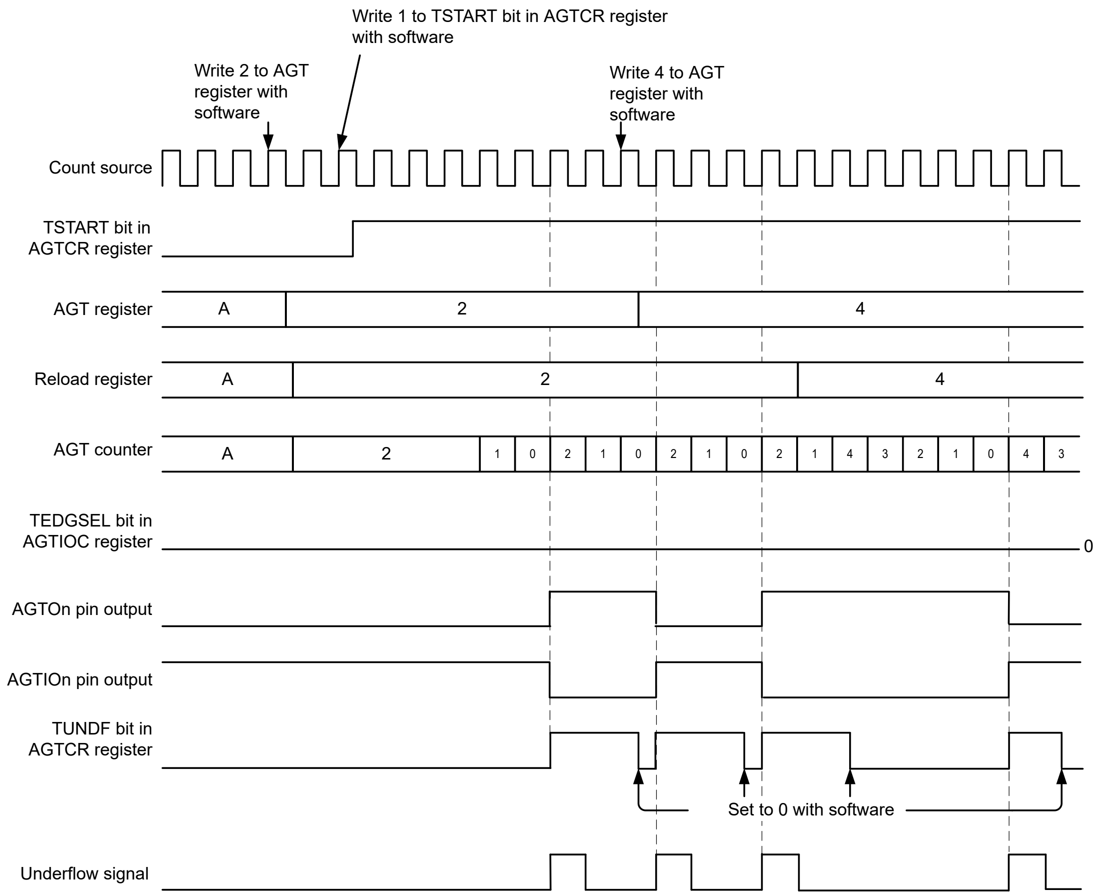


Figure 25.7 Operation example in pulse output mode

25.3.5 Event Counter Mode

In event counter mode, the counter is decremented by an external event signal (count source) input to the AGTIOn pin. Various periods for counting events can be set with the TIOGT[1:0] bits in the AGTIOC register and AGTISR registers. In addition, the filter function for the AGTIOn pin input can be specified with bits TIPF[1:0] in the AGTIOC register. The output from the AGTOn pin can be toggled even in event counter mode.

Figure 25.8 shows the operation example in event counter mode.

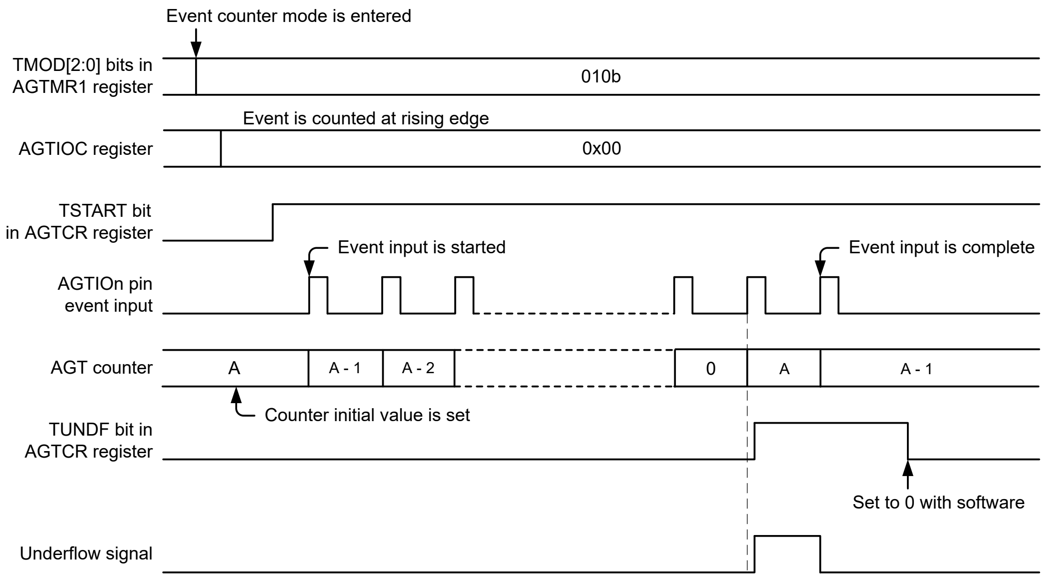
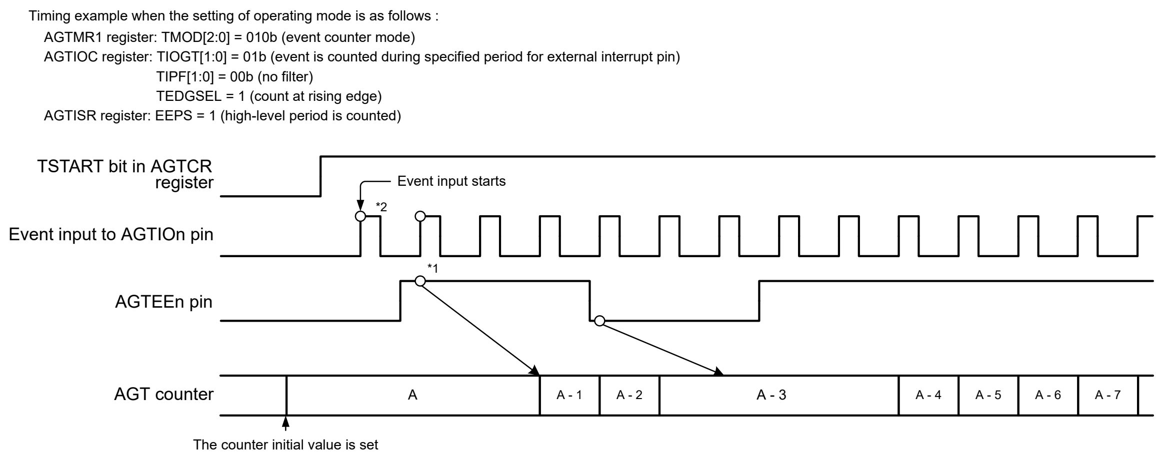


Figure 25.8 Operation example 1 in event counter mode

Figure 25.9 shows an operation example for counting during the specified period in event counter mode (TIOGT[1:0] bits in the AGTIOC register are set to 01b).



Note 1. To control synchronization, there is a delay of 2 cycles of the count source until the count operation is affected. It is also possible that the count start timing is shifted by 1 cycle because of the phase difference between the AGTEEn pin and the sampling clock.

Note 2. Count operation can be performed for 2 cycles of the count source immediately after the count starts, depending on the previous state before the count stops.
To disable the count for 2 cycles immediately after the count starts, write 1 to the TSTOP bit in the AGTCR register to initialize the internal circuit, and then complete the operation settings before starting the count operation.

Figure 25.9 Operation example 2 in event counter mode

25.3.6 Pulse Width Measurement Mode

In pulse width measurement mode, the pulse width of an external signal input to the AGTIOn pin is measured. When the level specified by the TEDGSEL bit in the AGTIOC register is input to the AGTIOn pin, the counter is decremented by the count source selected with the TCK[2:0] bits in the AGTMR1 register. When the specified level on the AGTIOn pin ends, the counter is stopped, the TEDGF flag in the AGTCR register is set to 1 (active edge received), and an interrupt request is generated. The measurement of pulse width data is performed by reading the count value while the counter is stopped. Also, when the counter underflows during measurement, the TUNDF flag in the AGTCR register is set to 1 and an interrupt request is generated.

Figure 25.10 shows the operation example in pulse width measurement mode.

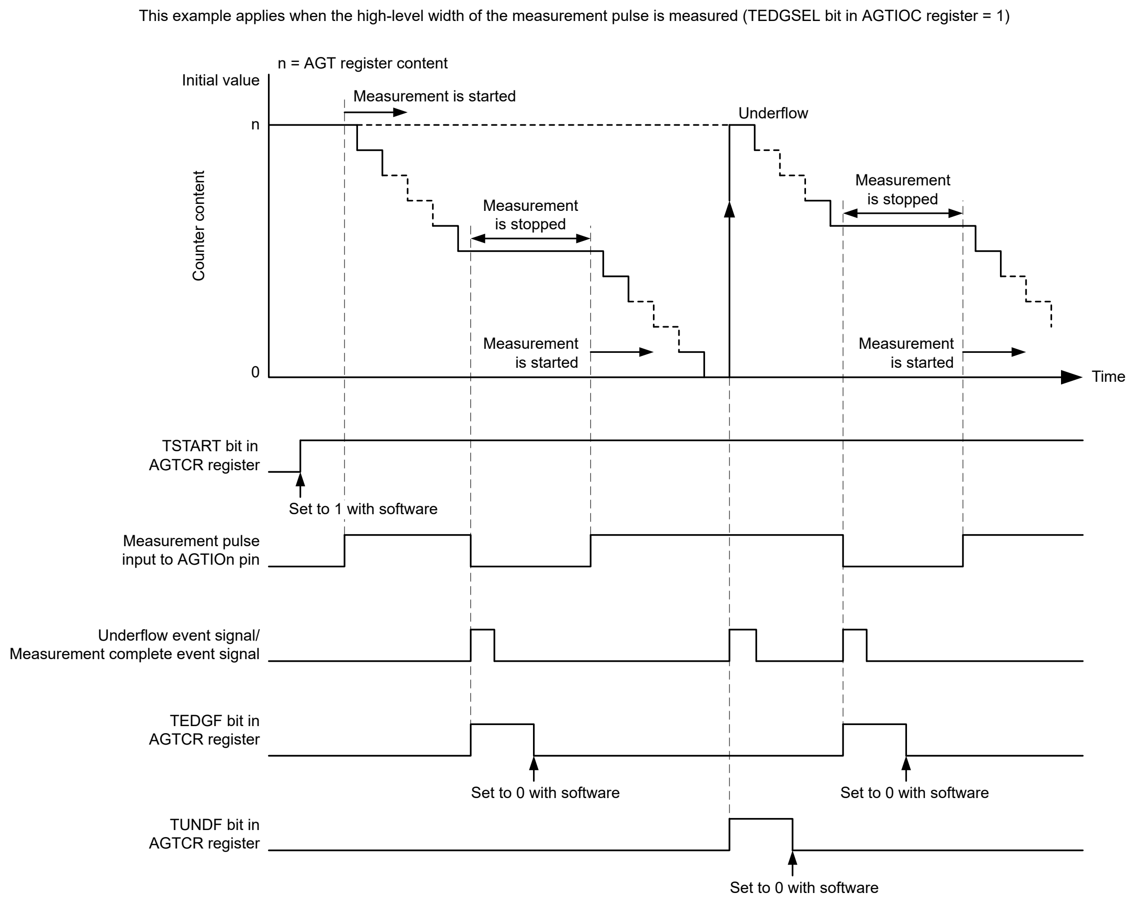


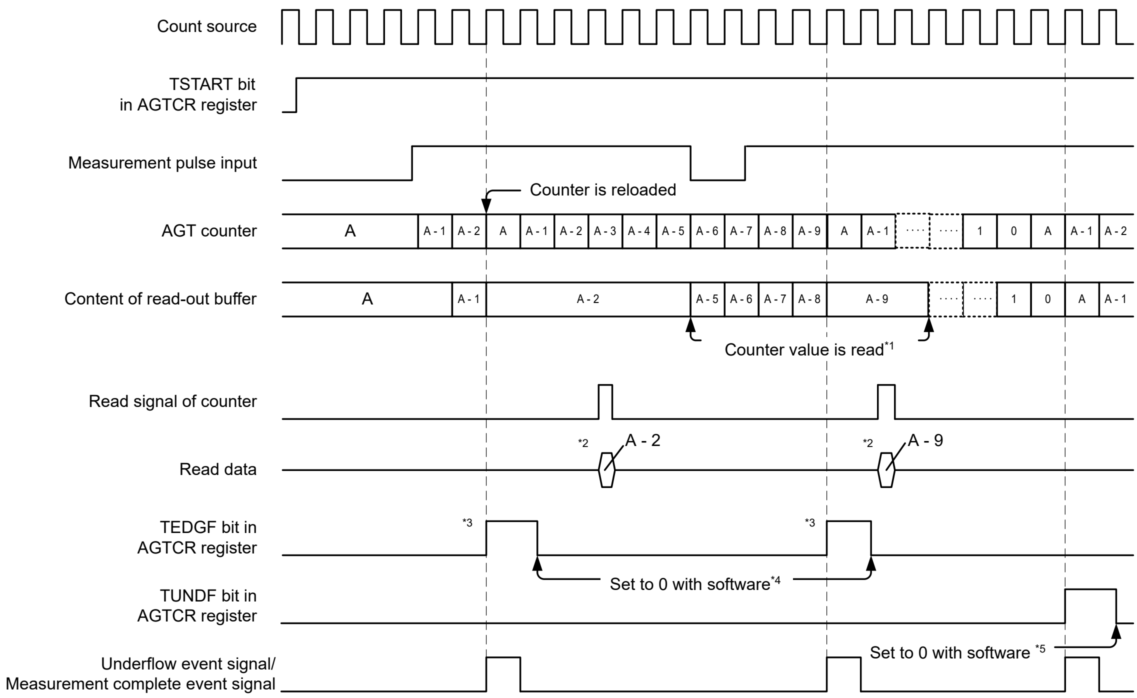
Figure 25.10 Operation example in pulse width measurement mode

25.3.7 Pulse Period Measurement Mode

In pulse period measurement mode, the pulse period of an external signal input to the AGTIOn pin is measured. The counter is decremented by the count source selected with TCK[2:0] bits in the AGTMR1 register. When a pulse with the period specified by the TEDGSEL bit in the AGTIOC register is input to the AGTIOn pin, the count value is transferred to the read-out buffer on the rising edge of the count source. The value in the reload register is loaded to the counter at the next rising edge. Simultaneously, the TEDGF flag in the AGTCR register is set to 1 (active edge received) and an interrupt request is generated. The read-out buffer (AGT register) is read at this time and the difference from the reload value (see section 25.4.6. How to Calculate Event Number, Pulse Width, and Pulse Period) is the period data of the input pulse. The period data is retained until the read-out buffer is read. When the counter underflows, the TUNDF flag in the AGTCR register is set to 1 (underflow) and an interrupt request is generated.

Figure 25.11 shows the operation example in pulse period measurement mode.

Only input pulses with a period longer than twice the period of the count source are measured. Also, the low-level and high-level widths must both be longer than the period of the count source. If a pulse period shorter than these conditions is input, the input might be ignored.



This example applies when the initial value of the AGT register is set to A, the TEDGSEL bit in the AGTIOC register is set to 0, and the period from one rising edge to the next edge of the measurement pulse is measured.

Note 1. Reading from the AGT register must be performed during the period from when the TEDGF flag is set to 1 (active edge received) until the next active edge is input. The content of the read-out buffer is retained until the AGT register is read. If it is not read before the active edge is input, the measurement result of the previous period is retained.
Note 2. When the AGT register is read in pulse period measurement mode, the content of the read-out buffer is read.
Note 3. When the active edge of the measurement pulse is input and then the set edge of an external pulse is input, the TEDGF flag in the AGTCR register is set to 1 (active edge received).
Note 4. To set to 0 with software, write 0 to the TEDGF flag in the AGTCR register with an 8-bit memory manipulation instruction.
Note 5. To set to 0 with software, write 0 to the TUNDF flag in the AGTCR register with an 8-bit memory manipulation instruction.

Figure 25.11 Operation example in pulse period measurement mode

25.3.8 Compare Match Function

The compare match function detects matches (compare match) between the content of the AGTCMA or AGTCMB register and the content of the AGT register. This function is enabled when the TCMEA or TCMEB bit in the AGTCMSR register is 1 (compare match A register or compare match B register is valid). The counter is decremented by the count source selected with the TCK[2:0] bits in the AGTMR1 register, and when the values of AGT and AGTCMA or AGTCMB match, the TCMAF/TCMBF flag in the AGTCR register is set to 1 (match), and an interrupt request is generated.

When the compare match function is enabled, the timing of the rewrite operation to the reload register and the counter differs. See section 25.3.1. Reload Register and Counter Rewrite Operation for details. In addition, the output level of the AGTOAn, AGTOBn pins is inverted by the match and by the underflow. The output level can be selected with the TOPOLA or TOPOLB bit in the AGTCMSR register.

Figure 25.12 shows the operation example in compare match function.

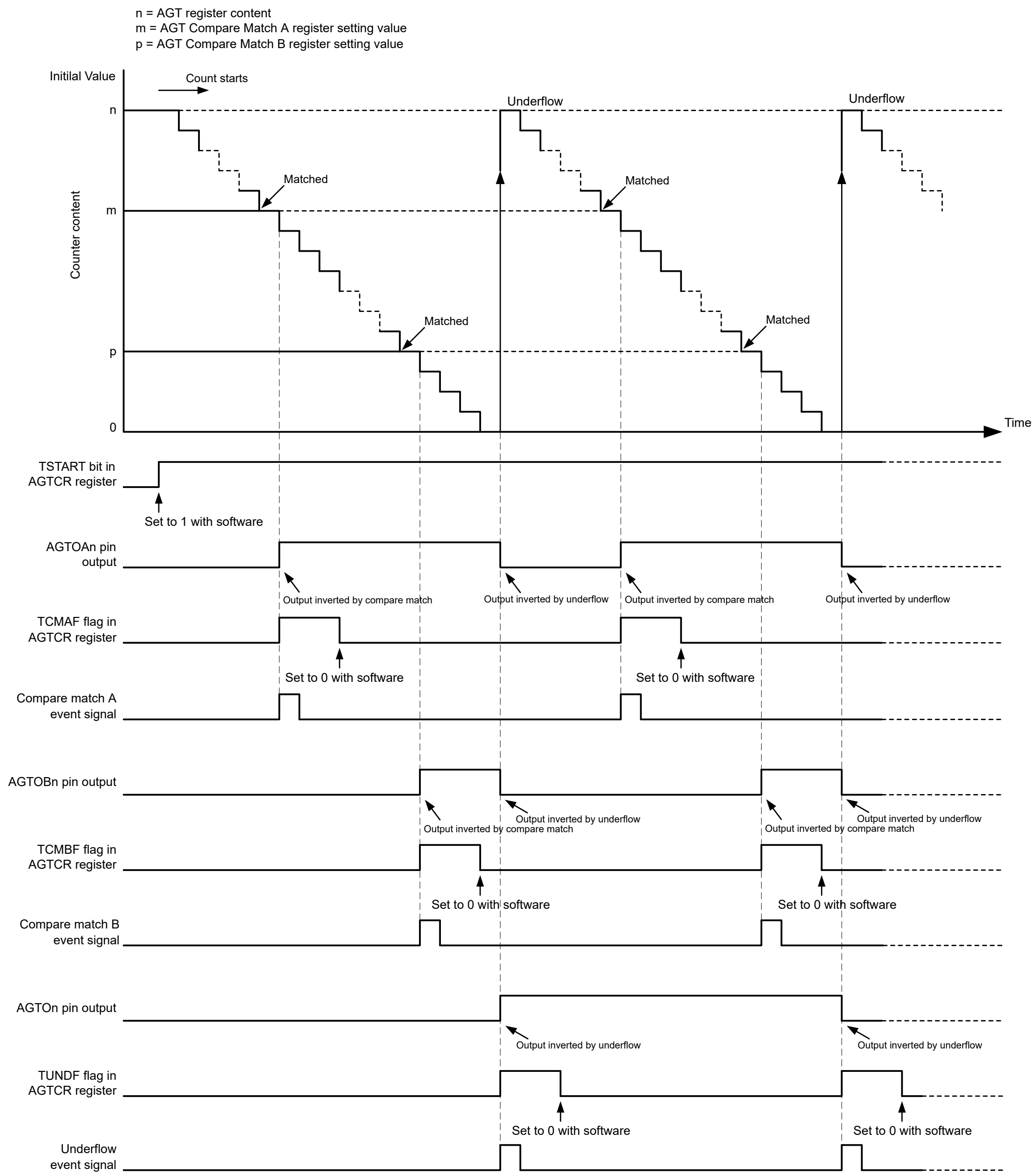


Figure 25.12 Operation example in compare match function (TOPOLA = 0, TOPOLB = 0)

25.3.9 Output Settings for Each Mode

Table 25.5 to Table 25.8 list the states of pins AGTOn, AGTIOn, AGTOAn, and AGTOBn pins in each mode.

Table 25.5 AGTOn pin setting

Operating mode	AGTIOC register		AGTOn pin output
	TOE bit	TEDGSEL bit	
All modes	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled

Table 25.6 AGTIOn pin setting

Operating mode	AGTIOC register TEDGSEL bit	AGTIOn pin I/O
Timer mode	0 or 1	Input (not used)
Pulse output mode	1	Normal output
	0	Inverted output
Event counter mode	0 or 1	Input
Pulse width measurement mode		
Pulse period measurement mode		

Table 25.7 AGTOAn pin setting

Operating mode	AGTCMSR register		AGTOAn pin output
	TOEA bit	TOPOLA bit	
Timer mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse output mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Event counter mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse width measurement mode	0	0	Prohibited
Pulse period measurement mode			

Table 25.8 AGTOBn pin setting

Operating mode	AGTCMSR register		AGTOBn pin output
	TOEB bit	TOPOLB bit	
Timer mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse output mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Event counter mode	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled (not used)
Pulse width measurement mode	0	0	Prohibited
Pulse period measurement mode			

25.3.10 Standby Mode

The AGT can operate in Software Standby mode. Set it to Software Standby mode with count operation start (TSTART = 1, TCSTF = 1).

Table 25.9 and Table 25.10 show the setting that can be used in Software Standby mode.

Table 25.9 Usable settings in Software Standby mode (AGT0)

Operating mode	AGTMR1.TCK[2:0]	Operating clock	Resurgence factor of CPU
Timer mode	100b or 110b	AGTLCLK or AGTSCLK	—
Pulse output mode	100b or 110b	AGTLCLK or AGTSCLK	—
Event counter mode*2 *3	—	AGTIO0*1	—
Pulse width measurement mode	100b or 110b	AGTLCLK or AGTSCLK	—
Pulse period measurement mode	100b or 110b	AGTLCLK or AGTSCLK	—

Note: —: invalid
Note 1. When using the AGTIO0 pin for external event input in Software Standby mode, set AGTIOSEL.TIES = 1.
Note 2. AGTEE pin is not available during Software Standby mode. External events are always enabled.
Note 3. When external event input is disabled during Software Standby mode (AGTIOSEL.TIES = 0), stop the count operation before the MCU enters Software Standby mode.
Restart the count operation after the MCU returns from Software Standby mode.

Table 25.10 Usable settings in Software Standby mode (AGT1)

Operating mode	AGTMR1.TCK[2:0]	Operating clock	Resurgence factor of CPU
Timer mode	100b or 110b or 101b*1	AGTLCLK or AGTSCLK or AGT0 underflow	• Underflow • Compare match A/B
Pulse output mode	100b or 110b or 101b*1	AGTLCLK or AGTSCLK or AGT0 underflow	• Underflow • Compare match A/B
Event counter mode*3 *4	—	AGTIO1*2	• Underflow • Compare match A/B
Pulse width measurement mode	100b or 110b or 101b*1	AGTLCLK or AGTSCLK or AGT0 underflow	• Underflow • Active edge
Pulse period measurement mode	100b or 110b or 101b*1	AGTLCLK or AGTSCLK or AGT0 underflow	• Underflow • Active edge

Note: —: invalid
Note: Release of Software Standby mode is only AGT1.
Note: Compare match A/B is resurgence factor of CPU from Software Standby mode.
Note 1. Only when AGT0 operates in Table 25.9
Note 2. When using the AGTIO1 pin for external event input in Software Standby mode, set AGTIOSEL.TIES = 1.
Note 3. AGTEE pin is not available during Software Standby mode. External events are always enabled
Note 4. When external event input is disabled during Software Standby mode (AGTIOSEL.TIES = 0), stop the count operation before the MCU enters Software Standby mode.
Restart the count operation after the MCU returns from Software Standby mode.

25.3.11 Interrupt Sources

The AGTn has three interrupt sources as listed in Table 25.11.

Table 25.11 AGT interrupt sources

Name	Interrupt source	DMAC/DTC activation
AGTn_AGTI	• When the counter underflows • When measurement of the active width of the external input pin (AGTIOn) is complete in pulse width measurement mode • When the set edge of the external input pin (AGTIOn) is input in pulse period measurement mode.	Possible
AGTn_AGTCMAI	• When the values of AGT register and AGTCMA register match	Possible
AGTn_AGTCMBI	• When the values of AGT register and AGTCMB register match	Possible

Note: Channel number (n = 0, 1)

25.3.12 Event Signal Output to ELC

The AGT uses the Event Link Controller (ELC) to perform a link operation to a specified module using the interrupt request signal as the event signal. The AGT outputs compare match A, compare match B, and underflow/measurement complete signals as event signals. For details, see section 20, Event Link Controller (ELC).

25.4 Usage Notes

25.4.1 Count Operation Start and Stop Control

- When the operating mode (see Table 25.1) is set to other than the event counter mode, or the count source is set to other than AGTn underflow event signal (TCK[2:0] = 101b):
  - After 1 (count starts) is written to the TSTART bit in the AGTCR register while the count is stopped, the TCSTF flag in the AGTCR register remains 0 (count stops) for 3 cycles of the count source. Do not access the registers associated with AGT other than the TCSTF flag until this flag is set to 1 (count in progress).
  - After 0 (count stops) is written to the TSTART bit during a count operation, the TCSTF flag remains 1 for 3 cycles of the count source. When the TCSTF flag is set to 0, the count is stopped. Do not access the registers associated with AGT other than the TCSTF flag until this flag is set to 0.
- When the operating mode (see Table 25.1) is set to event counter mode, or the count source is set to AGT1 underflow event signal (TCK[2:0] = 101b):
  - After 1 (count starts) is written to the TSTART bit in the AGTCR register while the count is stopped, the TCSTF flag in the AGTCR register remains 0 (count stops) for 2 PCLKB cycles. Do not access the registers associated with AGT other than the TCSTF flag until this flag is set to 1 (count in progress).
  - After 0 (count stops) is written to the TSTART bit during a count operation, the TCSTF flag remains 1 for 2 PCLKB cycles. When the TCSTF flag is set to 0, the count is stopped. Do not access the registers associated with AGT other than the TCSTF flag until this flag is set to 0.

25.4.2 Access to Counter Register

When the TSTART bit and TCSTF flag in the AGTCR register are both 1 (count starts), allow at least 3 cycles of the count source clock between writes when writing to the AGT register successively.

25.4.3 When Changing Mode

The registers associated with AGT operating mode (AGTMR1, AGTMR2, AGTIOC, AGTISR, and AGTCMSR) can be changed only when the count is stopped with both the TSTART bit and TCSTF flag set to 0 (count stops). Do not change these registers during count operation.

When the registers associated with AGT operating mode are changed, the values of TEDGF, TUNDF, TCMAF, and TCMBF flags are undefined. Before starting the count, write 0 to the following flags:

- TEDGF (no active edge received)
- TUNDF (no underflow)
- TCMAF (no match)
- TCMBF (no match).

25.4.4 Output Pin Setting

When using the AGTOn, AGTIOn, AGTOAn, or AGTOBn as an output pin, set up the Operation and determine the initial output values. Then set PmnPFS.PMR bit to 1.

When using the AGTIOn as an input pin in pulse width measurement mode or pulse period measurement mode, set up the Operation and start count operation. Then start to enter external events from the AGTIOn pin. Invalidate the first measurement and validate the second and later completed measurements.

25.4.5 Digital Filter

When using the digital filter, do not start the timer operation for 5 cycles of the digital filter clock after setting TIPF[1:0] bits and when the TEDGSEL bit in the AGTIOC register changes.

25.4.6 How to Calculate Event Number, Pulse Width, and Pulse Period

- In event counter mode, event number is expressed mathematically as follows:
  Event number = initial value of counter [AGT register] - counter value of active event end

- In pulse width measurement mode, pulse width is expressed mathematically as follows:
  Pulse width = counter value of stopping measurement - counter value of next stopping measurement
- In pulse period measurement mode, input pulse period is expressed mathematically as follows:
  Period of input pulse = (initial value of counter [AGT register] - reading value of the read-out buffer) + 1.

25.4.7 When Count is Forcibly Stopped by TSTOP Bit

After the counter is forcibly stopped by the TSTOP bit in the AGTCR register, do not access the following I/O registers for 1 cycle of the count source:

- AGT
- AGTCMA
- AGTCMB
- AGTCR
- AGTMR1
- AGTMR2.

25.4.8 When Selecting AGT0 Underflow as the Count Source

Operate according to the following procedures described in this section when selecting the underflow event signal as the count source.

(1) Procedure for starting operation

1. Set AGT.
2. Start the count operation of AGT1.
3. Start the count operation of AGT0.

(2) Procedure for stopping operation

1. Stop the count operation of AGT0.
2. Stop the count operation of AGT1.
3. Stop the count source clock of AGT1 (write 000b in the AGTMR1.TCK[2:0] bits).

25.4.9 Module-stop Function

AGT operation can be disabled or enabled using Module Stop Control Register D (MSTPCRD). The AGT module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode

25.4.10 When Switching Source Clock

When switching a clock source by changing SCKSCR.CKSEL[2:0], the clock output from the selector stops for 4 cycles of the switched clock. Therefore, when using the AGTIOn, AGTEEn, or both input as external event input, the clock source should not be switched. If switching the clock source while using the external event input, extend the input pulse width by 4 clock cycles of the switched source clock cycles.